

Physically-bounded litter-input priors

Literature basis and design choices for the `sigma_input` re-parameterisation (HIKET)

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1 Summary

`sigma_input` multiplies the litter forcing every model integrates. Under the current homogenised prior it is a weak nuisance parameter, and the likelihood drags two models (TP2, TP3) to effective litter fluxes of 34–50 tC/ha/yr — above the net primary production of *any* forest biome on Earth. This note records the literature basis for replacing that weak prior with a **physical bound on the effective litter flux**, homogeneous across all six models, and the design choices taken. It feeds the six `Prior_specs/*_priors.R` and the manuscript methods.

Decision: bound the effective flux to the *widest defensible boreal* window, $\text{sigma_input} \times J \in [0.5, 8.7]$ tC/ha/yr, anchored on the Gower et al. (2001) boreal total-NPP range.

2 The problem

The effective input flux is $\text{sigma_input} \times J$, where J is the tree-litter object shared by all six models (`Data/LitterData/tree_litter_per_site_year_awen*.csv`, Tupek et al.), median ≈ 2.8 tC/ha/yr across plots (`nwl + fw1 + cwl`: non-woody + fine-woody + coarse-woody litter, in AWEN pools).

The current prior treats `sigma_input` as a weak Tier-3 nuisance (log-transform, centre 1, log-SD 0.5 $\Rightarrow$ 95% CI $\approx [0.37, 2.66]$). But litter input has a physical referent the weak prior discards, so the posterior blows through it (Table 1; `doublechecks/effective_litter_flux_vs_physical.R`).

Table 1: Posterior-median effective litter flux by model.

Model	<code>sigma_input</code> (median)	effective flux (tC/ha/yr)	plausible?
SP1	0.32	0.8	yes
Yasso07	1.4	3.5	yes
Yasso15	1.9	4.8	yes
Yasso20	3.0	7.5	yes (very productive edge)

Model	σ_{input} (median)	effective flux (tC/ha/yr)	plausible?
TP2	13.6	34	no — 4× the most productive boreal forest
TP3	~20	50	no — 6×

TP2/TP3 buy their (best-in-study) R^2 by manufacturing carbon — the same root cause as the TP3 interannual oscillation (a cascade-starved fast pool force-fed to reach the observed stock). σ_{input} 's distance from ~ 1 is really a **structural-adequacy gauge**, interpretable only once the prior encodes the physical ceiling.

3 What σ_{input} legitimately represents

J is **tree litter only** (foliage/needles + fine roots + branches + stumps). Two physical terms are *not* in J and are what σ_{input} should absorb:

1. **Understory / ground-vegetation (incl. moss) litter** — genuinely missing from J . In Finnish boreal forest this is \$15–35% of total litter production (larger at northern, less productive sites): e.g. in a boreal *Betula pendula* stand total soil-C input (tree + understory) was 2.83 tC/ha/yr with understory **35%** $\approx$ **0.99 tC/ha/yr**. So $\sigma_{\text{input}} \approx 1.2\text{--}1.5$ is *physically justified by understory alone*.
2. **Tree-litter-model bias** — Tupek model error in the tree flux itself.

Neither mechanism can plausibly inflate soil input several-fold: a boreal forest's *total* detritus input to soil is capped by its net primary production.

4 Literature envelope (boreal, widest defensible)

The quantity to bound is **total forest detritus input to soil** (foliage + woody + fine roots + understory), i.e. the effective flux — **not** aboveground litterfall, which most papers report and which is only $\sim$ half of the total.

Anchor — **Gower et al. (2001)**. Boreal total (above- + belowground) NPP ranges **52–868 gC m⁻² yr⁻¹**, mean **424** $\rightarrow$ **0.52–8.68 tC/ha/yr**, mean **4.24**. At steady state a forest's total litter/detritus input to soil $\approx$ its NPP (all production eventually becomes detritus). NPP is in fact a **generous upper bound** on soil input, because in managed Finnish forests harvested stemwood leaves as timber rather than litter — exactly what “as wide as defensible” calls for. Corroborated across the boreal/temperate/tropical global database of Luyssaert et al. (2007).

Cross-checks — all comfortably inside [0.52, 8.68]:

- Finnish *Pinus sylvestris* measured tree litter: aboveground 1.31–2.36 + belowground 0.70–0.91 = **2.0–3.3 tC/ha/yr** (belowground 23–37% of total) — matches our J median 2.8.
- + understory/moss \$0.5–1.0 $\rightarrow$ typical boreal *total* input $\approx$ **2.5–4.3 tC/ha/yr**.
- Productive Norway spruce aboveground litterfall up to \$4 tC/ha/yr (central Europe \$8Mgha⁻¹\$ biomass; Berg et al. synthesis).

- **Global context:** even the most productive *tropical* forests reach total NPP $\approx 12\text{--}15$ tC/ha/yr. TP2/TP3's 34–50 tC/ha/yr exceed the NPP of **any** forest biome on Earth by 2–4 $\times$ — categorically impossible, let alone for boreal.

5 Chosen bounds

Physical window on the effective total-input flux (homogeneous, all six models):

$$\text{sigma_input} \times J \in [0.5, 8.7] \text{ tC/ha/yr}$$

anchored on the Gower et al. (2001) boreal total-NPP range (widest defensible; $\text{NPP} \geq \text{actual soil input}$). At the plot-median $J \approx 2.8$ this maps to a multiplier window $\text{sigma_input} \in \approx [0.18, 3.1]$ ($\approx [0.2, 3.5]$ at the lower $J \approx 2.5$).

sigma_init is already physically behaved (posteriors 0.19–0.72, all < 1 — the post-exploitation recovery narrative) and needs only a light guardrail ($\text{sigma_init} \in [0.1, 1.5]$). The two couple through the 1917 pre-run flux ($J_{1917} = J \times \text{sigma_init} \times \text{sigma_input}$); with both individually bounded the pre-run flux stays inside the same physical window, so no explicit joint constraint is imposed (verify post-run).

5.1 Consequences per model

- **SP1, Yasso07, Yasso15** (0.8 / 3.5 / 4.8): well inside — untouched.
- **Yasso20** (7.5): **within** the widest boreal envelope ($7.5 < 8.7$). At the *widest defensible* bound Yasso20 is a very-productive-but-possible boreal forest, so it is deliberately **not** bound. That Yasso20 sits at the physical edge is itself a reportable result.
- **TP2, TP3** (34 / 50): **hard-bound** at the ceiling. This removes their escape hatch $\rightarrow$ forces the misfit onto the (now prior-pinned) rates or into a visible R^2 hit = exposes structural inadequacy instead of hiding it.

6 Implementation

Bound the **effective flux** (the physical quantity), not the abstract multiplier, using each plot's own J :

- **Truncated-lognormal on sigma_input:** keep the weakly-informative centre-1 shape (log-SD 0.5) but impose a **hard wall** so that $\text{sigma_input} \times \bar{J} \in [0.5, 8.7]$, using the plot-mean $\bar{J}$ for the truncation. Truncated-lognormal over a flat uniform: retains the mild pull to 1 (no artificial preference for sitting at the ceiling) and mixes better near the wall in DEzs.
- **sigma_init:** light hard bounds $[0.1, 1.5]$.
- **Homogeneous across all six** Prior_specs/*_priors.R — the intercomparison requires identical construction.

6.1 Caveats

- **Mixing.** A leveraged parameter (TP2/TP3 sigma_input) clamped near a boundary can mix worse; check R-hat/ESS for sigma_input specifically after the re-calibration.
- **Scalar vs per-plot.** sigma_input is one scalar; truncation uses the mean $\bar{J}$, so a few high- J plots may nudge just past 8.7 at the multiplier ceiling. Second-order relative to killing the 13–20 $\times$ inflation; noted in methods.

- **Re-calibration required.** Full six-model re-run (Roihu) + convergence check.

7 Decisions taken

1. Scope = **boreal**, at the **widest defensible** end (Gower 2001 NPP max), not European/global (a tropical ceiling would not usefully constrain even TP2; a boreal ceiling is the physically correct one for a Finnish application).
2. Bound the **effective flux**, not the multiplier, so it is homogeneous in physical units across models with differing J .
3. **Floor from literature** (0.5 tC/ha/yr, Gower min), not an “understory-only- adds $\Rightarrow \text{sigma_input} \geq 1$ ” argument — keeps SP1’s sub-1 fit admissible rather than forcing it to a floor.
4. Shape = **truncated-lognormal** (weak centre-1 + hard physical wall).
5. **Yasso20 stays free** (widest envelope; $7.5 < 8.7$) — only TP2/TP3 bind.

8 References

- Gower, S.T. et al. (2001). Net primary production and carbon allocation patterns of boreal forest ecosystems. *Ecological Applications* 11(5):1395–1411.
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